

Data Science Module 5 Unit 5 Live session

Learning outcomes:

LO9: Explain the concepts of simple and multiple linear regression.

LO10: Identify and implement simple and multiple linear regression models using Python and scikit-learn.

LO11: Interpret regression plots, residual plots, and distribution plots to evaluate model performance.

LO12: Interpret polynomial regression models and pipelines to assess their predictive accuracy.

LO13: Compare different models (SLR, MLR, polynomial fit) based on mean squared error (MSE) and R-squared values.

LO14: Examine how developed regression models can support the prediction of car or laptop prices.

LO15: Select appropriate model evaluation techniques to assess the performance of regression models.

LO16: Select appropriate regularization techniques and hyperparameter tuning methods to optimize regression model performance and prevent overfitting.

LO17: Apply regression analysis techniques to predict real-world outcomes using a given dataset, providing justification for the model development process.

In this live session, you will engage directly with your Tutor and peers to review the preceding activity on housing. The session is designed to help you consolidate your learning from the previous sections, including importing and wrangling data, performing exploratory data analysis, developing predictive models, and evaluating their performance. You are encouraged to share your work, ask questions, and reflect on your approach to this activity.

The Head Tutor will guide the session, walking through the key steps of the activity workflow and highlighting insights from each section. While there is no formal submission in this session, you can share screenshots of your outputs – charts, tables, or model results – to foster collaborative discussion and peer feedback. This is an opportunity to discuss the challenges you faced, explore alternative approaches, and deepen your understanding of predictive modeling for real-world housing data.

You are invited to contribute actively via the live chat, sharing your experiences, questions, and observations. The session aims to create an interactive and supportive environment where you can learn from your peers, receive guidance from the Tutor, and strengthen your data analysis and modeling skills. Prepare to engage openly, exchange ideas, and celebrate the progress you have made throughout the project.

You can join the session via the Zoom LTI in the navigation bar.

Reminder:

It is important to have earphones and a microphone (and quiet surroundings) to participate in the session. You are advised to log in to the session 10 minutes early to enable your audio and visual settings and to test your connection before the session begins.

For guidance on how best to conduct conversations and interactions during live sessions, please review the live session etiquette guide before the session.